Forces on two objects

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Learning target: Identify an equal-and-opposite interaction pair on different objects.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 11.*
6 m/s²; the mass doubles at the same net force. Keep the net force fixed, including resistance.

02. Worked example: explain the model
Equal opposite interaction arrows on different bodies; acceleration also depends on mass.

03. Name both forces
Hand on wall; wall on hand. Equal forces on different objects.

04. Heavy and light carts
Equal forces; lighter cart greater acceleration for same force.

05. Balanced or a pair?
No; both act on book. Actual partners are book on Earth and book on table.

06. Review: connect an earlier idea *Review: Unit 1, Day 2.*
Distance 18 m; displacement +2 m (right); final position 8 m.

07. Exit check
Ground pushes foot forward; acts on foot. Forward external force can accelerate person.

08. STRETCH: test the model
Any inverse mass ratio, e.g. 20 and 60 kg; lighter skater accelerates three times as much.

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01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 11.*
Net force 0.24 N; new a≈0.3 m/s². Extra resistance reduces net force and acceleration if driving force is unchanged.

02. Worked example: explain the model
Equal opposite interaction arrows on different bodies; acceleration also depends on mass.

03. Equal forces, different results
aA 12 m/s²; aB 3 m/s²; opposite directions.

04. Identify actual partners
Book-on-Earth and book-on-table. Weight and normal both act on book; pair forces act on different objects.

05. Horse plus cart
Internal horse cart forces cancel in total; external ground force on horse can exceed resistance. Need net external force.

06. Review: connect an earlier idea *Review: Unit 1, Day 2.*
Distance 21 m; displacement -3 m; final position +4 m.

07. Exit check
Equal opposite forces; smaller-mass craft larger a. Comparison assumes other forces negligible or compares contribution from this interaction only.

08. STRETCH: test the model
System acceleration 10/5=2 m/s². Contact on 3 kg cart is 6 N; reaction on 2 kg cart is 6 N opposite the applied 10 N.